

GHZ tensors in the closure of matching tensors

An explicit truncation construction places ternary GHZ tensors in the closure of the matching-tensor image at every even order, limiting a class of proof strategies.

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Status

This is an exact arbitrary-order theorem over $\mathbb{C}$ about the *closure* of the set of perfect-matching tensors. It is a **structural no-go for a whole class of proof routes**, not a counterexample and not a proof of the Krenn-Gu conjecture. It changes no existing claim. The global conjecture remains **UNRESOLVED**.

Informally: for every even $n \geq 4$ the ternary GHZ tensor is a *limit* of perfect-matching tensors T_W with all blocks monochromatic matrix units, even though (for $n \geq 6$) the conjecture says it is never *equal* to one. Consequently no polynomial identity, rank bound, flattening bound, or any other condition that is closed under limits of the tensor T_W alone can separate the GHZ tensor from the hafnian image. A proof using the excluded closed tensor-only routes must instead use additional information, such as the exact fibre $T_W = \Delta$, that is, structure of W itself (such as the exact zero columns of the killer theorem), and any numerical search whose cost tends to zero along diverging weights has found nothing.

Setting

For even n and blocks $W_{ij} \in \mathbb{C}^{(3 \times 3)} (i < j)$, write

$$\begin{aligned} T_W(a_1, \dots, a_n) &= \sum_{(\text{perfect matchings } M \text{ of } K_n)} \prod_{\{i,j\} \in M} W_{ij}[a_i, a_j], \\ \Delta_n(a) &= 1 \text{ if } a_1 = \dots = a_n, \text{ else } 0. \end{aligned}$$

Let $\Phi_n : W \rightarrow T_W$ be the (polynomial) matching-tensor map and $\text{Im}(\Phi_n) \subset (C^3)^{\otimes n}$ its image. For a graph G on the vertex set $[n]$ with a proper three-edge-colouring $\kappa : E(G) \rightarrow \{0,1,2\}$ and a function $\nu : E(G) \rightarrow \mathbb{Z}$, define the one-parameter family

$$\begin{aligned} W_e(\epsilon) &= \epsilon^{\nu(e)} e_{\kappa(e)} e_{\kappa(e)}^T & \text{for } e \in E(G), \\ W_e(\epsilon) &= 0 & \text{for } e \text{ not in } E(G). \end{aligned} \quad (1)$$

Every block is a monochromatic matrix unit, so a perfect matching M of G contributes only to the word χ_M in which every vertex receives the colour of its M -edge:

$$T(W(\epsilon))(a) = \sum_{M \in \text{PM}(G), \chi_M = a} \epsilon^{\nu(M)}, \quad \nu(M) = \sum_{e \in M} \nu(e). \quad (2)$$

Theorem

Theorem A (face criterion). Let G be a cubic graph on n vertices with a proper three-edge-colouring whose colour classes M_0, M_1, M_2 are perfect matchings. Suppose $\nu : E(G) \rightarrow \mathbb{Z}$ satisfies

$$\begin{aligned} \nu(M_c) &= 0 & \text{for } c = 0,1,2, \\ \nu(M) &\geq 1 & \text{for every other perfect matching } M \text{ of } G. \end{aligned} \quad (3)$$

Then the family (1) satisfies, identically in ϵ ,

$$T(W(\epsilon)) = \Delta_n + \epsilon R(\epsilon), \quad (4)$$

where $R(\epsilon)$ is a tensor with polynomial entries in ϵ supported on non-constant words. In particular $T(W(\epsilon)) \rightarrow \Delta_n$ as $\epsilon \rightarrow 0$, and Δ_n lies in the Euclidean closure, hence in the Zariski closure, of $\text{Im}(\Phi_n)$. Condition (3) holds for some ν if and only if $\{M_0, M_1, M_2\}$ is the vertex set of a face of the perfect-matching polytope of G .

Theorem B (existence for every even order). For every even $n \geq 4$ there is a cubic graph G_n on n vertices with a proper three-edge-colouring and an integer potential ν_n satisfying (3). Explicitly, $G_4 = K_4$ with its three perfect matchings as colour classes and $\nu_4 = 0$, and G_{n+2} is obtained from G_n by replacing one vertex by a triangle (truncation). Hence

$$\Delta_n \in \text{closure}(\text{Im}(\Phi_n)) \quad \text{for every even } n \geq 4. \quad (5)$$

Corollary C (no tensor-level separation). Let P be any polynomial on $(\mathbb{C}^3)^{\otimes n}$ with $P(T_W) = 0$ for all W , or more generally any subset $Z \subset (\mathbb{C}^3)^{\otimes n}$ closed in the Euclidean topology with $\text{Im}(\Phi_n) \subset Z$. Then $P(\Delta_n) = 0$, respectively $\Delta_n \in Z$. No such condition can exclude a Krenn–Gu witness. In particular, upper bounds on flattening ranks, slice ranks, border ranks, or on any other lower-semicontinuous invariant of T_W , and every identity in the ideal of the hafnian image, are automatically satisfied by Δ_n .

Corollary D (numerical searches). For every even $n \geq 4$ and every nonnegative continuous loss L on $(\mathbb{C}^3)^{\otimes n}$ with $L(0)=0$, the infimum over W of $L(T_W - \Delta_n)$ is zero. If moreover L is faithful, meaning $L(X)=0$ only for $X=0$ (for example any norm), then for $n \geq 6$ that infimum is attained only in the limit of unbounded weights: unconditionally at $n = 6$ by the six-vertex exclusion, and conditionally on the conjecture for $n \geq 8$. A cost tending to zero is therefore not evidence for a witness unless the weights stay bounded in some vertex gauge. (The faithfulness hypothesis was added after an external review noted that a non-faithful loss can vanish at a bounded non-witness.)

Proof

Theorem A

Let M be a perfect matching of G inducing a constant word c^n . Every edge of M then has colour c , so $M \subset M_c$; since both are perfect matchings, $M = M_c$. Conversely M_c induces c^n . Hence by (2)

$$T_W(\epsilon)(c^n) = \epsilon^{\nu(M_c)} = 1 \quad \text{for every } \epsilon,$$

using the first line of (3). Every other perfect matching induces a non-constant word and has $\nu(M) \geq 1$, so each non-constant coefficient $T_W(\epsilon)(a) = \sum \epsilon^{\nu(M)}$ is a polynomial in ϵ with zero constant term. This is (4), and the limit statement follows because the entries of R are polynomials. A Euclidean limit of points of $\text{Im}(\Phi_n)$ lies in the Zariski closure of $\text{Im}(\Phi_n)$ because polynomials are continuous.

For the face statement: the perfect-matching polytope of G is the convex hull of the incidence vectors of its perfect matchings, and these vectors are exactly its vertices. A set S of vertices of a polytope is the vertex set of a face if and only if some linear functional attains its minimum over the polytope

exactly on S . For $S = \{M_0, M_1, M_2\}$ such a functional, shifted to have minimum zero and scaled to be integral with gap at least one, is precisely a potential ν satisfying (3), and conversely. (The integrality scaling uses that a rational functional can be chosen, since the polytope is rational.)

Theorem B

Truncation. Let G be cubic with a proper three-edge-colouring κ whose classes are perfect matchings, let v be a vertex with incident edges $v u_c$ of colour c ($c=0,1,2$), and let ν satisfy (3) for G . Define G' by deleting v , adding three new vertices t_0, t_1, t_2 , and adding the edges

$u_c t_c$	with colour c ,	(attachment edges)
$t_a t_b$	with colour $3-a-b$, $a < b$.	(triangle edges)

At t_c the three incident colours are c , $3-c-a = b$, and $3-c-b = a$, so the colouring is proper, and every colour class $M'_c = M_c - v u_c$

- $u_c t_c + t_a t_b$ is again a perfect matching.

Perfect matchings of G' . A perfect matching of G' uses either one or three attachment edges: with none, the three triangle vertices cannot be matched among themselves; with two, the remaining triangle vertex has no free partner. If it uses exactly $u_c t_c$, it must contain $t_a t_b$, and its remaining edges form a perfect matching of $G - v - u_c$; adding $v u_c$ gives a perfect matching M of G , and this correspondence $M \rightarrow M - v u_c + u_c t_c + t_a t_b$ is a bijection between perfect matchings of G and perfect matchings of G' with one attachment edge. If it uses all three attachment edges, its remaining edges form a perfect matching N of $G - N[v]$, where $N[v] = \{v, u_0, u_1, u_2\}$; conversely every such N extends uniquely.

Potential. Fix an integer $A \geq 1$ and put

$\nu'(u_c t_c)$	$= \nu(v u_c) + A,$	
$\nu'(t_a t_b)$	$= -A,$	
$\nu'(e)$	$= \nu(e)$	for every other edge.

A lifted matching satisfies $\nu'(M') = \nu(M) - \nu(v u_c) + (\nu(v u_c) + A)$

- $A = \nu(M)$. Hence $\nu'(M'_c) = 0$ and every lifted extra matching keeps $\nu' \geq 1$. A three-attachment matching has

$$\nu' = 3A + \sum_c \nu(v u_c) + \nu(N) \geq 1$$

as soon as $A \geq (1 - \sum_c \nu(v_{u_c}) - \min_N \nu(N))/3$, where the minimum is over perfect matchings N of $G - N[v]$ (vacuous if there are none). Thus ν' satisfies (3) for G' .

Induction. K_4 with colour classes its three perfect matchings has no other perfect matching, so $\nu = 0$ satisfies (3). Truncating any vertex gives the triangular prism on six vertices, and iterating gives cubic graphs G_n for every even $n \geq 4$ together with potentials ν_n satisfying (3). Theorem A then gives (5).

The canonical family used by the verifier truncates the vertex of largest label at every step and takes the smallest admissible A . For the prism this is $A = 1$: the three rungs receive ϵ^1 , the three triangle edges opposite to them ϵ^{-1} , and the single extra matching (the three rungs) contributes ϵ^3 to one mixed word.

Corollaries C and D

Corollary C is the continuity statement already used in Theorem A. For Corollary D, $L(T_\epsilon(W)) - \Delta_n \rightarrow L(0) = 0$ along the family of Theorem B, which gives the infimum. For the second statement let L be faithful and let W_k be a sequence, bounded in some vertex gauge, with $L(T(W_k)) - \Delta_n \rightarrow 0$. A convergent subsequence has a limit W^* with $L(T(W^*)) - \Delta_n = 0$ by continuity of Φ_n and L , hence $T(W^*) = \Delta_n$ by faithfulness; at $n = 6$ this contradicts the six-vertex exclusion, and at general $n \geq 6$ it contradicts the conjecture.

Consequences for the programme

1. **Closed tensor-only separators are insufficient.** The conjecture is the statement $\Delta_n \notin \text{Im}(\Phi_n)$ for $n \geq 6$, while $\Delta_n \in \text{closure}(\text{Im}(\Phi_n))$ for all n . A proof cannot consist solely of closed conditions containing the whole image of $T_\epsilon W$; an available alternative is to derive structure of W from the exact equation, as the column-killer theorem, the diagonal-anchor theorem, and the Laplace/Wick identities do. Rank-drop, sensor-rank, and flattening arguments are valid only in the role they already play in the repository: as consequences of exact identities in W , never as conditions on T alone.
2. **The asymptotic witnesses are exactly Bogdanov-type structures.** Every limiting family in Theorem B is a cubic graph with three monochromatic perfect matchings whose extra matchings are suppressed by weights, not cancelled. For $n \geq 6$ the extra matchings always exist (the cubic diagonal exclusion), so ν is genuinely nonzero and the weights are unbounded in every vertex gauge. An exact witness would have to achieve with cancellation what these families achieve only in the limit.

3. **Numerical evidence.** Unconstrained complex Levenberg–Marquardt on the full $n=6$ and $n=8$ systems repeatedly drives the least-squares cost to 10^{-12} – 10^{-15} while the largest weights grow to 10^1 – 10^4 ; in every such run all nonzero blocks are monochromatic matrix units on a cubic three-edge-coloured skeleton, with weights spread over several orders of magnitude. These are the families of Theorem A, not witnesses. The repository’s exploratory optimizer should therefore be read with this theorem in mind.

Boundary

- The theorem says nothing about $\text{Im}(\Phi_n)$ itself; it does not exclude, construct, or approximate an exact witness.
- It does not claim that every point of the closure of $\text{Im}(\Phi_n)$ arises from a matrix-unit family, nor classify all faces of matching polytopes containing three colour classes.
- The $n = 6$ unconditional statement in Corollary D depends on the repository’s six-vertex computer-assisted exclusion; the $n \geq 8$ statement is conditional on the conjecture.
- The literature consulted (Krenn–Gu–Soltesz; Chandran–Gajjala; Chandran–Gajjala–Illickan, MFCS 2024; the maintained problem page) treats exact realizability only; no statement about closures or limiting families was found there, but this novelty assessment is bounded and not a literature search of record.

Verification

```
python claims/arbitrary-order/verify_ghz_closure_matching_polytope_face_asymptotic_realizability.py
python claims/arbitrary-order/audit_ghz_closure_matching_polytope_face_asymptotic_realizability.py
```

The primary verifier builds the canonical truncation family for $n = 4, \dots, 16$, enumerates all perfect matchings of each G_n , checks the proper colouring and the perfect-matching colour classes, checks (3) exactly in integers, and for $n \leq 10$ expands $T(W(\epsilon))$ symbolically over all perfect matchings of K_n to confirm (4) as a polynomial identity. The independent audit rebuilds the family with a different perfect-matching enumeration, recomputes the potentials, and evaluates the full 3^n tensor numerically at $\epsilon = 10^{-1}, 10^{-2}, 10^{-3}$ for $n = 6, 8, 10$, confirming that all constant coefficients equal one to machine precision and that the largest mixed coefficient scales like $\epsilon^{(\nu_{\min})}$ with the recorded minimum extra potential. The written proof above is the theorem; the programs are exact and numerical supporting checks of the displayed family.

Publication record

This edition was selected for Lyrebird on September 22, 2026. This edition explicitly requires a nonnegative continuous loss in Corollary D, as needed for its infimum claim, and limits the proof-route conclusion to closed tensor-only separators. Supporting programs were rerun for this release. These edits do not claim new independent peer review.

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